

HARSTENA, Sweden

WMO: 025630

Lat: 58.251N

Lon: 17.011E

Elev: 46

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.5	16.9	3.4	6.6	21.3	10.0	9.2	19.5	28.4	31.6	25.5	30.9	9.4	330	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	76.4	66.1	73.7	64.8	71.4	63.5	68.6	73.6	66.8	71.6	65.2	69.8	8.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	98.1	71.3	64.9	92.4	69.8	63.1	86.7	68.3	32.8	73.8	31.4	71.6	30.2	69.9	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.0	22.0	19.6	DB	9.5	81.4	6.6	2.7	4.7	83.4	0.9	85.0	-2.9	86.5	-7.7	88.5	(4)
(5)			WB	8.7	70.3	6.5	2.2	4.0	71.9	0.2	73.2	-3.4	74.4	-8.2	76.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.5	31.0	30.9	35.3	41.7	50.3	58.9	65.1	64.3	57.0	47.4	39.9	34.6
	DBStd	13.27	6.69	6.63	6.25	4.80	5.04	4.28	4.38	4.33	4.64	5.26	5.24	7.04
	HDD50	2789	590	534	455	251	59	1	0	0	5	112	305	477
	HDD65	6874	1055	954	920	698	456	188	54	65	242	546	754	942
	CDD50	1497	0	0	1	3	68	269	467	443	214	31	1	0
	CDD65	105	0	0	0	0	0	5	56	43	1	0	0	0
	CDH74	216	0	0	0	0	1	15	120	80	0	0	0	0
	CDH80	16	0	0	0	0	0	1	10	5	0	0	0	0
Wind	WSAvg	10.0	12.2	11.1	10.2	9.0	8.5	8.6	8.0	8.5	9.7	10.8	11.9	11.8
Precipitation	PrecAvg	21.9	1.5	1.2	1.1	1.2	1.6	2.0	2.5	2.4	2.0	2.0	2.1	1.9
	PrecMax	27.3	2.8	2.0	2.1	2.7	3.0	3.7	5.5	5.6	4.2	4.4	4.2	3.3
	PrecMin	17.8	0.2	0.6	0.4	0.2	0.7	0.6	0.5	0.0	0.7	0.1	0.4	0.7
	PrecStd	2.7	0.7	0.4	0.5	0.7	0.6	0.9	1.3	1.3	0.8	1.0	0.9	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.9	47.7	56.3	59.6	67.7	75.6	81.1	80.0	70.7	61.4	53.1	51.7
		MCWB	43.3	41.8	46.4	50.0	56.1	60.6	67.9	67.5	61.1	55.3	49.3	47.8
	2%	DB	43.2	44.0	50.2	54.9	63.6	71.4	77.1	76.0	67.8	58.2	50.6	47.5
		MCWB	40.1	39.8	42.6	46.6	54.1	58.7	67.0	66.3	60.4	53.8	47.7	44.0
	5%	DB	40.9	41.5	46.4	51.7	60.6	68.5	74.5	73.3	65.5	56.2	48.3	44.7
		MCWB	38.7	38.1	40.7	44.9	52.6	57.5	65.7	65.1	59.5	52.6	45.7	42.1
	10%	DB	39.1	39.1	43.3	49.0	58.0	66.0	72.3	70.9	63.3	54.4	46.3	42.6
		MCWB	37.3	36.7	38.9	43.4	51.4	56.7	64.7	63.8	58.4	51.4	44.3	40.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.8	43.1	47.6	51.7	58.2	63.8	71.5	71.7	63.7	57.2	50.7	48.0
		MCDB	46.2	46.5	55.2	58.4	63.9	72.0	76.3	76.5	67.9	59.8	52.0	51.4
	2%	WB	40.9	40.3	43.6	47.9	55.7	61.7	69.2	68.5	62.0	54.8	48.1	44.5
		MCDB	42.4	43.2	48.6	52.9	61.7	68.3	74.7	73.0	65.9	57.1	50.2	46.5
	5%	WB	39.0	38.5	41.3	45.7	53.9	60.1	67.5	66.7	60.6	53.4	46.2	42.6
		MCDB	40.5	40.9	45.6	50.6	59.4	66.0	72.7	71.1	64.0	55.6	47.8	44.2
	10%	WB	37.4	37.0	39.4	43.9	52.2	58.6	66.0	65.2	59.2	51.8	44.8	40.9
		MCDB	39.0	38.9	42.6	48.2	56.9	64.1	70.8	69.6	62.4	54.1	46.0	42.4
Mean Daily Temperature Range	5% DB	MDBR	6.3	7.0	9.0	10.3	10.9	11.4	11.2	10.4	8.7	6.9	5.6	6.1
		MCDBR	7.9	9.2	13.4	14.4	14.3	14.8	13.8	12.9	11.4	9.0	7.9	8.1
		MCWBR	6.9	6.7	8.5	8.3	7.8	7.4	6.8	6.8	6.8	6.3	7.1	7.1
	5% WB	MCDBR	7.2	8.4	12.5	13.7	13.4	13.1	11.8	11.0	10.4	8.2	7.1	7.4
		MCWBR	6.6	6.5	8.4	8.6	8.3	7.6	6.3	6.9	6.9	6.3	6.7	6.8
Clear-Sky Solar Irradiance	taub	0.292	0.325	0.341	0.379	0.375	0.364	0.394	0.384	0.353	0.342	0.318	0.279	
	taud	2.272	2.324	2.366	2.322	2.408	2.473	2.410	2.429	2.496	2.424	2.291	2.200	
	Ebn at Noon	175	223	255	262	272	277	265	259	251	216	160	137	
	Edh at Noon	15	22	28	34	33	32	33	30	25	20	15	11	
All-Sky Solar Radiation	RadAvg	108	290	766	1282	1707	1874	1712	1367	918	409	142	76	
	RadStd	15	54	73	129	133	83	155	122	108	62	30	10	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)		+0.70	N/A	N/A	N/A	N/A	N/A	-220	-283	N/A	N/A

Nomenclature: See separate page